

Mini Project Report - Secure Basic Website

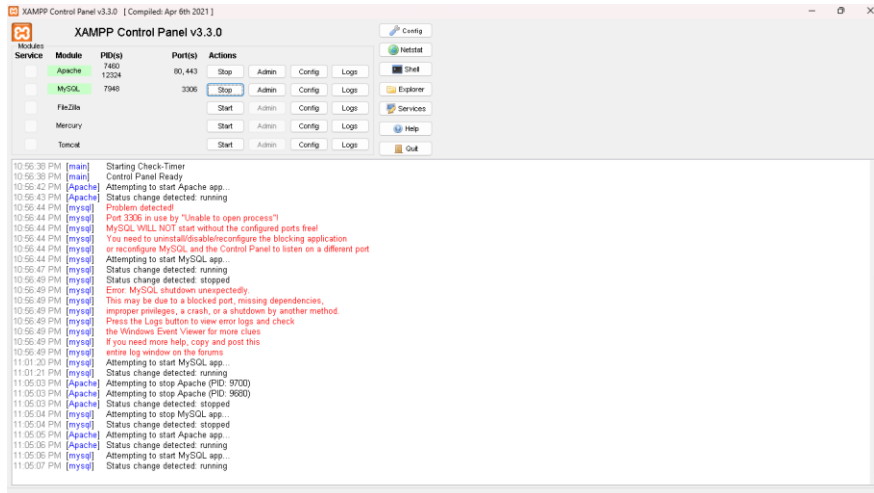
Introduction:

The goal of this project is to develop and secure a simple website by implementing fundamental security practices. This ensures the website is protected from common vulnerabilities such as cross-site scripting (XSS) attacks, insecure data transmission, and improper input handling. The project demonstrates the importance of basic security measures to create a safer web environment.

Features Implemented:

- 1. **Input Validation:**
 - o Ensured all user inputs are validated both on the client and server sides.
 - o Implemented rules to accept only the expected data format, reducing risks of injection attacks.
- 2. **HTTPS Enforcement:**
 - o Configured the web server to redirect all HTTP traffic to HTTPS, ensuring encrypted communication.
 - o Implemented SSL/TLS certificates to secure data transmission.
- 3. **XSS Prevention:**
 - o Sanitized user inputs to prevent malicious scripts from being executed.
 - o Applied Content Security Policy (CSP) headers to restrict the execution of untrusted scripts.

Name	Date modified	Type	Size
 index	2/1/2025 11:15 PM	Chrome HTML Do...	1 KB
 index.html	2/1/2025 11:02 PM	Text Document	1 KB
 process.php	2/1/2025 11:03 PM	Text Document	1 KB
 process	2/1/2025 11:10 PM	PHP Source File	1 KB



Login Form

Username:

Pass: Please match the requested format.

Login Form

Username:

Password:

```
<?php
// Force HTTPS
if (empty($_SERVER['HTTPS']) || $_SERVER['HTTPS'] === "off") {
    $redirect = "https://" . $_SERVER['HTTP_HOST'] . $_SERVER['REQUEST_URI'];
    header("Location: $redirect");
    exit();
}

// Check if the form is submitted
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    // Secure Input
    $username = htmlspecialchars(trim($_POST['username']));
    $password = htmlspecialchars(trim($_POST['password']));

    // Validate Username (only letters and numbers)
    if (!preg_match("/^[A-Za-z0-9]{4,12}$/", $username)) {
        die("Invalid username format.");
    }

    // Validate Password Length
    if (strlen($password) < 6) {
        die("Password must be at least 6 characters.");
    }

    // Dummy Login Check
    if ($username == "user123" && $password == "pass1234") {
        echo "Welcome, " . htmlspecialchars($username);
    } else {
        echo "Invalid login credentials.";
    }
}
?>
```